

Not all LED Drivers Provide 'Clean' Light

LED drivers with a poor power factor reflect harmonics back to the mains. As the power consumption of high-power LEDs used as light sources increases erratically, there is a risk of 'dirty electricity' across the mains. To get to the root of this problem, high-power LED drivers must come with power factor correction



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Incandescent bulbs are essentially ohmic resistors and consume sinusoidal current from the mains grid. The power factor of these devices is essentially '1.' With LED lights, things are somewhat more complex.

LEDs are semiconductors operated with direct current. Their characteristic shows a marked kink at approximately 3V. When the maximum value is ex-

ceeded, the LED might be destroyed. LEDs therefore require special drivers that convert the mains voltage to a constant direct current. This constant current ensures that all LEDs in a chain are lit at equal brightness—irrespective of the threshold voltage.

Such LED drivers are, however, not ohmic resistors, but instead consumers with a power factor that tends to be far below '1.' This leads to the reflection of harmonics back to the mains, resulting

in undesired reactive currents.

Pulsed direct current causes problems

The above problem results from the need to convert alternating current into constant direct current. To do this, the current must be rectified and stabilised by a capacitor with sufficient capacitance. The capacitor is charged through the half wave to its peak value and supplies energy until the next half wave reaches the capacitor value.

If the voltage at the rectifier is greater than that from the capacitor, a brief high-amplitude current is generated during the respective half wave. This current peak is much higher than would be expected based on the power rating. The resulting current is no longer sinusoidal and includes a large share of harmonics (the steeper the edge, the higher the harmonic share). This problem arises from the fact that the alternating current needs to be rectified at the input and smoothed before it can be used further down the line. If a converter is installed to generate the required constant current from the high direct voltage, the situation becomes even worse.

Pulse-width modulation corrects power factor

Since it is expected that LED lighting systems will replace other lighting solutions across the board, corrective measures must be taken in order to ensure that the mains quality does not deteriorate too much. EN 61000-3-2 standard therefore demands that LED drivers rated 25W and higher come